

Smoke Testing

A short set of critical checks used to decide whether a build is stable enough for further testing.

What is Smoke Testing?

Smoke Testing is a short and objective verification of the most critical paths of a build or environment. Its purpose is not to test everything in depth, but to answer quickly: “is this version stable enough for us to continue testing?”. If essential functions fail, running extensive regression can waste time and produce results with little value.

When to execute

Execute after a deployment, new build, important environment update or before larger cycles such as regression and UAT. In pipelines, an automated smoke suite can also work as an initial gate before more expensive suites.

How to define the scope

Choose a few high-value, high-impact scenarios. In e-commerce: accessible Home, search, PDP, add to cart, authentication and start of checkout. In a banking system: login, balance inquiry and a critical transaction. The scope should reflect the business, not a generic list.

How to execute in practice

Prepare the environment and minimum data, execute critical paths quickly and record only the evidence needed to make a decision. The objective is to detect blockers, unavailability, configuration errors, broken essential integrations and failures that prevent subsequent testing.

Practical example

After an e-commerce deployment: 1) open Home; 2) search for a product; 3) open PDP; 4) add to cart; 5) start checkout. If the cart returns error 500, smoke fails and the team evaluates blocking regression until the problem is understood.

Smoke is not regression

Smoke verifies minimum stability with a few critical scenarios. Regression looks for side effects across much broader coverage. An approved smoke does not mean the product is defect-free; it only means there is minimum confidence to continue.

Approval criteria

Define beforehand which failures block the cycle. A cosmetic failure may not prevent continuation; unavailable login in a product that requires authentication probably does. The decision should consider risk and impact.

Automation

Smoke is an excellent candidate for automation when flows are stable and repetitive. Keep the suite short, reliable and fast; a smoke suite that takes hours loses part of its purpose.

Quick checklist

Correct build; available environment; authentication; main navigation; critical transaction; indispensable integration; minimum data; blocking errors; evidence of the decision to proceed or stop.

QA Mindset

Smoke does not try to prove that everything works. It quickly looks for evidence that it is worth investing in the remaining tests.